Nagaraj B

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Education

Indian Institute of Information Technology, Guwahati Guwahati, Assam

Bachelor of Technology in Computer Science and Engineering — CGPA: 7.01 *2023 – 2027*

Available for internships from January 2027

Technical Skills

Languages: C++, SQL, Python, JavaScript, Java

Backend & APIs: Node.js, Express, FastAPI, REST APIs

AI & ML: Scikit-learn, Pandas, NumPy, Regression, Classification, Feature Engineering, Model Evaluation

Frontend & Mobile: React.js, HTML, CSS, Kotlin, Android

Databases: MongoDB, MySQL, Redis

Tools: Git, GitHub, Docker, Linux

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Software Engineering

Projects

KVix — In-Memory Key-Value Server | *C++, epoll, RESP* GitHub

- Built a single-threaded key-value server in C++ using `epoll`, implementing 11 commands through a Redis RESP subset compatible with `redis-cli` and `redis-benchmark`.
- Handled partial and pipelined commands, measuring about 173.6K GET requests/sec at 0.12 ms latency with 50 concurrent connections in a small-store benchmark.
- Implemented durable writes by syncing an append-only log before acknowledging each write, with snapshots and log replay for crash recovery.
- Implemented LRU eviction using a hash map and doubly linked list, plus TTL expiration; verified recovery of keys and their expiry times after forced process termination.

Where Is My Bus — Delhi Bus Journey Planner | *Flutter, Fastify/TypeScript, PostGIS, SQLite* GitHub

- Built an offline-first bus journey-planning app for Delhi, allowing users to search stops, routes and direct buses without internet access.
- Developed a GPS-processing pipeline that matched bus locations to routes and filtered invalid readings using speed and direction checks.
- Estimated bus arrival ranges using historical travel patterns and separate weekday/weekend models, rather than showing a single arrival time.
- Implemented simulation to cross-check reported bus GPS locations using a trusted authority and simulated satellite timing measurements.

Saarthi — Gig Worker App | *Kotlin, Android, Node.js, Express, Python, FastAPI, MongoDB* GitHub

- Built a voice-first Android app in Kotlin for Indian gig workers, backed by Node.js/Express and Python FastAPI services, with check-ins in 14 languages via Google speech and translation.
- Designed a structured six-step check-in workflow with input validation and retry paths, keeping voice conversations consistent instead of relying on free-form LLM responses.
- Trained XGBoost models for earnings prediction and burnout-risk classification on 50,000 synthetic records, with rule-based fallbacks when the ML service was unavailable.
- Built a background notification engine with quiet hours, cooldowns and daily limits to prevent repeated alerts, backed by MongoDB and Jest tests.

Achievements & Extracurriculars

- Solved 600+ DSA problems across platforms (Codolio), including 400+ on LeetCode.
- 3-star (rating 1715) on CodeChef; rating 1319 on Codeforces.
- Semi-finalist, Flipkart GRiD 8.0.